

First Analyze Then Enhance: A Task-Aware System for Speech Separation, Denoising, and Dereverberation

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Abstract

This paper presents the *First Analyze Then Enhance (FATE)* framework for speech enhancement. In FATE, observed signals are initially classified based on their specific degradation types and then enhanced using appropriate modules tailored to each type. This design prevents the overprocessing of clean signals and reduces process complexity by eliminating unnecessary procedures. This paper focuses on commonly encountered degradations, including additive noise, reverberation, speech mixing, and their combinations. To address these degradations, FATE includes a separation submodule and a denoising/dereverberation submodule the enhancement models. Degradation types are predicted using features extracted from pretrained models based on automatic speech recognition and self-supervised learning. Experiments show that FATE can accurately identify undegraded signals and achieve comparable enhancement performance for degraded signals in each scenario while optimizing processing complexity.

Index Terms: Speech enhancement, First Analyze Then Enhance, Single-channel, Speech separation

1. Introduction

Speech degradation, manifesting as audio signal distortions, is a common challenge in real-world applications. These degradations encompass but are not limited to, noisy speech (background noise) [1], reverberant speech (echoes from reflections) [2], and mixed speech (overlapping voices) [3]. Traditionally, these degradations have been addressed independently using specialized methodologies tailored to their distinct objectives. For example, speech separation is particularly challenging because individual speaker information must be conserved, which introduces a permutation problem and influences both the model structure and training objectives.

Recently, powerful models [4, 5, 6, 7] have blurred the boundaries between tasks by simultaneously purifying and separating signals. This trend encourages tackling more comprehensive tasks using unified models, as exemplified by the UR-GENT Challenge [8]. Employing a single model enables to use of diverse datasets, leading to broader knowledge acquisition and more powerful models. However, this approach increases training difficulty and demands greater computational resources (e.g., more data and larger networks) to achieve sufficient generalization. Furthermore, the internal structure of unified models is often opaque, making it difficult to apply task-specific losses to enhance performance on individual tasks. Therefore, it is crucial to carefully select the tasks to merge, ensuring they can be trained using shared loss function. This consideration has led to the adoption of multistage approaches [9, 10, 11], in which different types of degradation are addressed using specialized

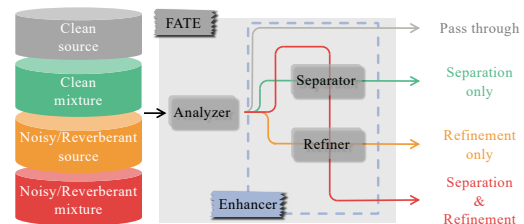


Figure 1: Overview of the FATE framework.

modules. The multistage structure also offers flexibility in incorporating different network architectures, enabling improved performance on individual tasks.

This paper proposes a framework capable of handling multiple degradations in a multistage manner. Unlike conventional approaches, which pass observed signals through all submodules, we argue that the type of data degradation should be identified before processing. Each degradation is then directed to a corresponding tailored module based on the classification. We refer to our method as *First Analyze Then Enhance (FATE)*. This design allows undegraded signals to pass through the system while reducing processing complexity by skipping unnecessary submodules. We first evaluate the framework on commonly encountered tasks, including denoising, dereverberation, and separation. Notably, the framework is flexible and can be extended to other enhancement tasks, such as declipping and inpainting. In a related study, a similar framework was proposed that combined an analyzer and a multistage processor to remove audio effects [12]. The key differences lie in the target signal and main motivation, as we focus on speech signals and aim to reduce the overprocessing of undegraded inputs. Unlike audio effects, speech signals present challenges in distinguishing between several types of degradation due to their similar characteristics, such as reverberation and multi-speaker scenarios.

As shown in Fig. 1, FATE comprises two main modules: an analyzer and an enhancer. The analyzer must be highly accurate, as misclassification can lead to incorrect routing and directly affect performance. We utilize models pretrained based on automatic speech recognition and self-supervised learning [13] as the backbone of the analyzer due to their powerful feature extraction capabilities. Observed signals are categorized into four groups, based on whether they require separation or refinement (i.e., denoising/dereverberation). Therefore, the enhancer includes a separator and a refiner. We incorporate an encoder-decoder-attractor (EDA) mechanism to estimate the number of speakers [14], addressing the challenge of an unknown speaker count. In addition, the enhancer is designed with a separation-priority pipeline (SPP) [10] and is trained in multiple phases to ensure that it enhances the observed signals using only the necessary submodules with sufficient performance.

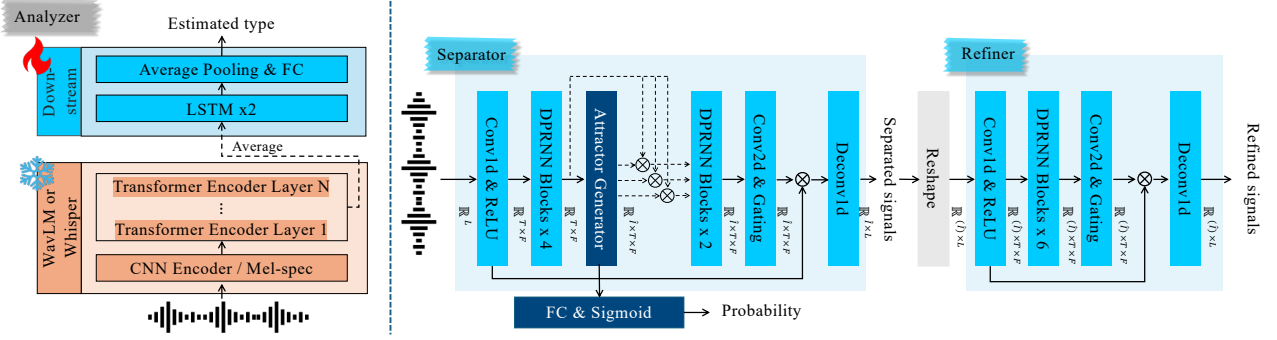


Figure 2: Left: Architecture of the analyzer, where the backbone model is frozen during fine-tuning. Right: Architecture of the separator with EDA and refiner, where dark blue and dotted lines indicate the inner procedures of the EDA mechanism. Note that “Reshape” is performed in the route including both submodules.

2. FATE

2.1. Research scope

FATE aims to address a broad range of speech degradations using an assembled system in which submodules can be flexibly added, removed, or combined for different applications while maintaining a low fine-tuning cost. In this study, we explore three fundamental combinations: speech separation, denoising, and dereverberation. This results in four data categories, each with an adaptive processing objective, summarized as follows: (C1) **clean source**: passes through; (C2) **clean mixture**: separation only; (C3) **noisy/reverberant source**: refinement only; (C4) **noisy/reverberant mixture**: separation and refinement.

2.2. Framework details

In FATE, the observed signals are first classified into predefined categories by the analyzer module and then processed according to routing based on the predicted category.

The degradation-type analyzer is built on pretrained models, leveraging their rich feature representations learned from large-scale data. We adopt Whisper [15] and WavLM [16] because of their robustness to noise and their ability to capture both acoustic and contextual information. Acoustic features are expected to help distinguish noisy and clean signals, whereas contextual features are expected to aid in differentiating reverberant and mixed signals. Following SUPERB’s principle [17] that downstream models should remain simple, we adopt a lightweight architecture comprising two long short-term memory (LSTM) layers, a time-dimensional average pooling layer, and a fully connected (FC) layer. As illustrated on the left-hand side of Fig. 2, the model aggregates features across different transformer encoder layers and uses them to classify the observed signals into four categories.

The enhancer consists of a separator and a refiner. The separator addresses mixed signals, whereas the refiner handles noise and reverberation. The image on the right side of Fig. 2 illustrates the route of a signal $X \in \mathbb{R}^L$ that requires both separation and refinement, with architectures based on a dual-path recurrent neural network (DPRNN) [18], where L denotes the signal length in the time domain. The DPRNN forms a latent representation of an input signal $\mathbb{R}^{T \times F}$ through a 1D convolutional layer followed by an activation function (e.g., ReLU), where T and F denote the frame and feature dimensions, respectively. The input feature is fed into the DPRNN blocks, where intra-

and inter-chunk operations are performed iteratively and alternately. Separation masks for individual speakers are then estimated using a WaveNet-like gating mechanism equipped with convolutional layers [19]. The separated signals are obtained by masking the input features and performing overlap-add in the 1D deconvolution. The separator and refiner submodules operate within a cascaded SPP [10] structure. The separated signals are independently processed by the refiner by folding the source dimension into the batch dimension. This design effectively manages complex tasks while maintaining compatibility with simpler subtask processing.

To handle an arbitrary number of speakers in mixed signals, the separator is integrated with an LSTM-based encoder-decoder attractor (EDA) [14]. The EDA module estimates the number of speakers $\hat{I}$ in the input and produces the corresponding speaker representations for subsequent separation. During training, given an I -speaker mixture, the EDA module recurrently generates $\hat{I} + 1$ speaker attractors and their existence probabilities $\hat{p}_i$. Generation stops when the probability falls below a threshold, and the first $\hat{I}$ attractors are used.

2.3. Training strategies of FATE

We train the model in multiple stages using the following strategies to ensure that each module performs well, both individually and in combination. **Noise-Adaptation Training (NAT)**: The EDA may struggle to accurately predict the number of speakers in noisy and reverberant conditions. To fully leverage its potential, EDA-equipped models are pre-trained on noise-free mixtures and then fine-tuned on noisy-reverberant signals containing single and multiple speakers [5]. This approach is referred to as noise-adaptation training (NAT). **Divide-and-Integrate Training (DIT)**: The enhancer is trained using the divide-and-integrate (DIT) method. Specifically, the separator is trained on data requiring separation, and the refiner is trained on data requiring refinement. Subsequently, the enhancer is assembled and trained on a full degraded dataset. The first stage ensures the performance of each submodule when operating individually. The subsequent stage improves performance when the modules are combined by mitigating the impact of information loss from preceding submodules. **NAT&DIT**: We adopt a three-stage training process to train FATE with a separator incorporating the EDA mechanism. In the first two stages, the separator is trained using NAT, while the refiner is trained conventionally. In the third stage, both submodules are fine-tuned together.

Table 1: Reverberation configuration.

Room	Length (m)	$\mathcal{U}(5, 10)$
	Width (m)	$\mathcal{U}(5, 10)$
	Hight (m)	$\mathcal{U}(3, 4)$
T60	Time (s)	$\mathcal{U}(0.2, 0.8)$
Receiver	Length (m)	$\frac{L_{room}}{2} + \mathcal{U}(-0.2, 0.2)$
	Width (m)	$\frac{W_{room}}{2} + \mathcal{U}(-0.2, 0.2)$
	Hight (m)	$\mathcal{U}(0.9, 1.8)$
Sources	Hight (m)	$\mathcal{U}(0.9, 1.8)$
	Distance (m)	$\mathcal{U}(0.66, 2)$
	angular	$\mathcal{U}(0, 2\pi)$

Table 2: Eleven data types and corresponding categories (Cat.) and the stages at which they were used for training. Checkmark “✓” denotes that data is used for entire model training or fine-tuning, while “sep.” and “ref.” indicate that the data were used for the corresponding submodule training.

Cat.	Data type	NAT		DIT		NAT & DIT		
		1st	2nd	1st	2nd	1st	2nd	3rd
C1	Clean 1-src							
C2	Clean 2-src	✓		sep.	✓	sep.		✓
	Clean 3-src							
C3	Noisy 1-src		✓	ref.	✓	ref.		✓
	Reverb. noisy 1-src							
C4	Noisy 2-src							
	Noisy 3-src							
	Reverb. 2-src		✓	sep.	✓		sep.	✓
	Reverb. 3-src							
	Reverb. noisy 2-src							
	Reverb. noisy 3-src							

2.4. Loss functions

The analyzer is trained using four-class cross-entropy:

$$\mathcal{L}_{\text{analy.}} = - \sum_c e_c \log \hat{e}_c. \quad (1)$$

where $\hat{e}_c$ and e_c denote the estimated probability and ground truth of the c -th data type, respectively. Following the predominant approach in discriminative speech separation research, we employ the negative scale-invariant signal-to-noise ratio (SI-SNR) [20] as the loss function for the refiner, and its permutation-invariant training (PIT) [21] version for the separator, which are expressed as

$$\mathcal{L}_{\text{ref.}} = -\text{SI-SNR}(\hat{s}, s), \quad (2)$$

$$\mathcal{L}_{\text{sep.}} = -\max_{\pi \in \mathcal{P}} \frac{1}{I} \sum_i \text{SI-SNR}(\hat{s}_{\pi(i)}, s_i), \quad (3)$$

where π and $\mathcal{P}$ represent a permutation order and the set of all permutations, respectively. For EDA-equipped models, binary cross-entropy (BCE) loss is used alongside the SI-SNR loss, given as:

$$\mathcal{L}_{\text{EDA}} = \frac{1}{I+1} \sum_{1 \leq i \leq I+1} \text{BCE}(\hat{p}_i, p_i), \quad (4)$$

where $p_1, \dots, p_I$ are set to ones, denoting the existence of the first I speakers, and p_{I+1} is set to zero as a stop flag.

3. Experiments

3.1. Dataset

We followed the Libri-3Mix scripts [22] to generate datasets with training and validation sets containing 33,901 and 3,000 utterances for each condition, respectively. 2-source and 1-source signals were generated by removing 1 or 2 source signals before mixing. Noise was sampled from WHAM! [23], and reverberation was simulated following WHAMR! [24] with pa-

Table 3: Overall quantitative results averaged across ten types (excluding clean 1-src) achieved by each model with their parameter numbers (Param.) of the enhancer. The first row in FATE shows the results obtained using the analyzer, whereas the second row represents the upper bound with 100% classifier accuracy. Bold font indicates the highest score.

Methods (Number of Blocks)	Training manner	SI-SNR (dB)	STOI (%)	PESQ	Param. (M)
Unprocessed	-	-4.22	61.01	1.44	-
DPRNN (6)	E2E	3.77	74.21	1.82	3.7
DPRNN (12)	E2E	4.88	76.78	1.94	7.2
DPRNN (6)	NAT	5.47	77.98	2.01	3.7
DPRNN (12)	NAT	5.77	78.65	2.05	7.2
DPRNN_EDA (6)	NAT	5.79	79.01	2.04	4.0
DPRNN_EDA (12)	NAT	6.35	80.20	2.12	7.5
FATE (12)	DIT	6.67 (6.68)	80.48 (80.46)	2.13 (2.13)	7.3
	NAT & DIT	7.02 (7.03)	80.95 (80.92)	2.17 (2.17)	7.3
FATE_EDA (12)	NAT & DIT	7.18 (7.22)	81.61 (81.60)	2.18 (2.19)	7.6

rameters listed in Table 1. This resulted in 11 types of observed signals, summarized in the second column of Table 2. The entire dataset was unbalanced, with the (C4) noisy/reverberant mixture being three times larger than the size of other categories used for training. Dynamic mixing was applied during training and validation. For testing, the entire test dataset was used to generate 3,000 utterances for each type of degradation, producing a total of 33,000 utterances.

3.2. Model parameters and training details

We used WavLM-base¹, WavLM-large², and Whisper³, which have 12, 24, and 12 transformer encoder layers respectively, as the backbones of the analyzer [15, 16]. The two LSTM layers had output dimensions of 64, and were mapped to four neurons via an FC layer. We fine-tuned the analyzer with a batch size of 24, ensuring that each category contained six samples. The Adam optimizer was used with an initial learning rate of 0.0001, which remained constant for the first 20 epochs. Afterward, the learning rate was annealed by a factor of 0.98 every two epochs. The model was trained for a total of 150 epochs.

Each separator and refiner contained 6 DPRNN blocks. For comparison, we used DPRNN with 6 and 12 blocks as baselines. The DPRNN block took input features generated by a 1D convolution with a window length of 16, hop size of 8, and 64 channels. During internal local and global processing, the data chunk length and hop size were set to 250 and 125, respectively. The number of output channels for separation models without EDA was set to 3, corresponding to the largest number of speakers in the experiment. The batch size was 16. The initial learning rates for pre-training and fine-tuning were 0.001 and 0.0001, respectively. The learning rate was halved if the model failed to improve validation performance for five consecutive epochs. Training continued for a total of 150 epochs, with early stopping triggered after 30 epochs without improvement. We investigated the training strategies described in Subsection 2.3 as well as end-to-end (E2E) training, where the entire dataset was used in a single-stage training. The subdatasets used for

¹<https://huggingface.co/patrickvonplaten/wavlm-libri-clean-100h-base-plus>

²<https://huggingface.co/patrickvonplaten/wavlm-libri-clean-100h-large>

³<https://github.com/openai/whisper-small>

Table 4: Data type-specific performance in terms of SI-SNR [dB], STOI [%], and PESQ. Bold font indicates the best score.

Data type	Unprocessed			DPRNN (NAT)			DPRNN_EDA (NAT)			FATE (NAT & DIT)			FATE_EDA(NAT & DIT)		
	SI-SNR	STOI	PESQ	SI-SNR	STOI	PESQ	SI-SNR	STOI	PESQ	SI-SNR	STOI	PESQ	SI-SNR	STOI	PESQ
Clean 1-src	93.31	99.99	4.55	25.37	98.76	4.23	12.33	85.01	3.15	As unprocessed			As unprocessed		
Clean 2-src	0.00	70.80	1.62	13.65	91.65	2.81	13.84	92.07	2.89	14.71	92.43	2.89	14.93	92.75	2.92
Clean 3-src	-3.70	58.85	1.38	8.22	82.06	2.05	9.35	84.91	2.25	8.79	83.16	2.13	9.34	84.68	2.22
Noisy 1-src	1.01	74.91	1.64	11.92	88.42	2.52	12.09	88.67	2.57	12.43	89.23	2.62	12.37	89.13	2.59
Reverb. noisy 1-src	-2.95	68.85	1.57	4.67	82.90	2.17	4.87	83.07	2.19	5.88	84.56	2.29	5.84	84.46	2.27
Noisy 2-src	-3.06	62.08	1.42	9.08	82.77	2.13	9.40	83.63	2.20	10.28	85.46	2.32	10.22	85.48	2.33
Noisy 3-src	-5.28	54.79	1.34	5.49	74.57	1.79	6.70	77.92	1.93	7.21	79.10	1.99	7.35	79.70	2.02
Reverb. 2-src	-5.29	61.08	1.45	3.38	79.16	2.03	3.66	79.95	2.07	5.00	81.47	2.18	4.99	81.82	2.17
Reverb. 3-src	-7.86	52.04	1.32	0.16	67.77	1.63	1.10	70.67	1.71	1.68	70.52	1.72	2.00	72.51	1.76
Reverb. noisy 2-src	-6.53	56.59	1.37	2.08	73.40	1.79	2.46	74.44	1.82	3.69	76.58	1.91	3.57	76.57	1.90
Reverb. noisy 3-src	-8.58	50.14	1.31	-0.92	63.80	1.54	0.07	66.63	1.60	0.53	66.96	1.62	1.21	69.01	1.66

each stage-varying strategy are summarized in Table 2. For DPRNN and DPRNN_EDA, the loss was calculated between the output signals and clean target sources. For FATE, the target signal s_i was clean during the first stage of training and noisy or noisy-reverberant during the second stage. These implementations were carried out using Asteroid toolkit⁴ [25].

4. Results and discussions

4.1. Overall quantitative results

In addition to SI-SNR [20], we employed short-time objective intelligibility (STOI) [26] and perceptual evaluation of speech quality (PESQ) [27] as evaluation metrics. For the EDA-equipped models, we followed the procedure described in the Subsection 3.3 of [14] to address the mismatch between the estimated number of speakers and the ground truth. For models without EDA, we used the combination that achieved the maximum SI-SNR relative to the ground truth. The overall quantitative results are presented in Table 3. Here, we report the results of FATE using a WavLM-large-based analyzer owing to its superior performance. Both FATE and FATE_EDA demonstrated superior performance compared to the baseline models. Specifically, FATE achieved an improvement of 1.25 dB, and FATE_EDA achieved an improvement of 0.83 dB over DPRNN and DPRNN_EDA, respectively, when using the same number of blocks. Given that the upper bound of FATE was even higher, these results suggest that task-specific modules have the potential to outperform unified models. Notably, the model size remains nearly unchanged, further underscoring the advantages of FATE. Comparing models trained with and without NAT, we found that NAT contributed to improvements of 0.89 dB in SI-SNR for DPRNN with 12 blocks and 0.35 dB for FATE, indicating the effectiveness of NAT in enhancing performance.

4.2. Data-type-specific performance analysis

Table 4 presents data-type-specific results for the best system in each group. All models evaluated consisted of 12 DPRNN blocks. First, we compared the results for clean single-source signals. FATE successfully processed all undegraded signals by passing them through the system, whereas the unified models resulted in severe distortions. When comparing DPRNN and FATE, FATE showed a greater advantage in handling complex tasks, particularly data from ‘C4’. FATE achieved an SI-SNR improvement of over 1.5 dB in ‘C4’, compared to improvements of 0.8 dB in ‘C2’ and ‘C3’. One important reason for this is the SPP design, which provides a dedicated route for processing complex data in FATE, maximizing the benefits of

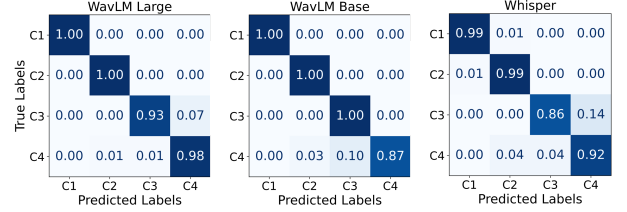


Figure 3: Confusion matrices of the analyzers using WavLM-Large, WavLM-Base, and Whisper. ‘C1’, ‘C2’, ‘C3’, and ‘C4’ indicate clean source, clean mixture, noisy/reverberant source, and noisy/reverberant mixture, respectively. The numbers represent the ratio of each case to the total number of labels.

multitask learning while efficiently mitigating oversuppression. When integrating EDA into FATE, the advantages of FATE, such as improved perceptual quality and higher SI-SNR, were preserved. In addition, the capability to handle simple tasks was enhanced. Further investigation showed that the EDA module struggles with complex data, such as noisy-reverberant mixed signals, due to its difficulty in accurately predicting the number of speakers, with an accuracy between 70-80%, which limits separation performance.

4.3. Performance of analyzer

Fig. 3 shows the confusion matrices of the analyzers with different pretrained models. The clean single-source and clean mixture are two classes that can be easily classified, with WavLM-based models achieving 100% accuracy and the Whisper-based model achieving 99% accuracy. All models struggled to distinguish between degraded single-source and degraded mixtures due to noise and reverberation. On average, WavLM-large achieved the highest accuracy and best performance and was therefore used to guide FATE. Future work will include reducing the analyzer’s model size using techniques such as knowledge distillation to achieve a more efficient system.

5. Summary

Instead of integrating speech separation, denoising, and dereverberation into a unified model, this paper has proposed the FATE framework, which first analyzes the type of data degradation and then enhances it according to the identified needs. By carefully designing the submodules in sequence following the SPP, and employing training strategies such as NAT and DIT, FATE outperformed unified models in the aforementioned tasks, demonstrating the effectiveness of combining task-specific models.

⁴<https://github.com/asteroid-team/asteroid>

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